Foundational Requirements

The following conditions must be in place before any AI use case can be effectively implemented. These requirements apply across all capability dimensions.

1. **Data Accessibility**: All relevant data sources — internal systems, editorial databases, and external datasets — must be retrievable and accessible to the implementation team.
2. **System Interoperability**: Data sources must be connectable within a shared environment. Siloed systems that cannot exchange data programmatically will block cross-source modelling regardless of data availability.

Investigative Reporting

Using AI-driven analytical tools to process large volumes of unstructured data, uncover trends and anomalies, and develop impactful news stories.

Prerequisites

1. **Existing Articles & Internal Archives**
    a. Full text of past investigative reports, editorial notes, published articles, and associated metadata
2. **Public Records & Structured Data**
    a. Government registries, corporate filings, court rulings, open-data portals, and statistical datasets
3. **Image & Media Data**
    a. Photographs and video footage for identity verification and visual evidence
4. **Domain Expert Access**
    a. Subject-matter experts available for fact identification, hypothesis formation, and validation

Modelling Processes & Model Outputs

Use Case	Required Data	Modelling Process	Model Output	Reference Case
Image-Based Identity Verification	2 + 3	Facial recognition, image matching, identity cross-referencing against public records	Identity match reports with confidence scores	Ukrainian Children Investigation
Data-Driven Investigative Leads	1 + 2 + 4	Entity extraction, association and anomaly detection, cross-language document matching	Entity relationship networks and anomaly reports	AI-Powered Reporting on New Legislators

Technical Assessments

Data collection: When no ready-made data source exists, web scraping is required. Modern websites commonly employ anti-scraping mechanisms; manageable at small scale, but a data provider may be needed for larger volumes.

Data processing: Data arriving in different formats requires corresponding specialised tools for cleaning and transformation.

Data exploration: Using AI to explore data requires guarding against hallucinations; an evidence-based approach must be adopted to verify factual sources.

Cross-language and cross-source processing: Matching documents across languages (e.g., Chinese and English company registrations, foreign-language court rulings) demands attention to proper-noun consistency and translation quality.

Sensitive data handling: Data must be de-identified before being sent to an LLM, or a local model should be used instead. Compliance with Taiwan's Personal Data Protection Act must also be ensured.

Interactive Data Visualization

Using AI to analyse and synthesise large datasets, enhancing visual storytelling to create more engaging content that reaches a broader audience.

Prerequisites

1. Article Content & Metadata
    a. Full text of past articles, headlines, tags, publication date, author, category
2. External Structured Data
    a. Government open data, statistical datasets, geographic and regional data
3. External Unstructured Data
    a. Social media posts, third-party data providers
4. Domain Expert Access
    a. Subject matter experts available for feature definition and model validation

Modelling Processes & Model Outputs

Use Case	Required Data	Modelling Process	Model Output	Reference Case
Data-Driven Visual Storytelling	1 + 2	Structured data cleaning, feature extraction, interactive chart generation	Embeddable interactive data visualisations	Ebola Virus Hotspot Map
Sentiment Analysis Visualisation	1 + 3	Spam filtering, full-article semantic encoding, sentiment classification	Sentiment trend charts by political figure	Election Social Media Sentiment Analysis
Predictive Visual Insights	1 + 2 + 4	Feature engineering, prediction models (logistic regression / tree-based), expert validation	Predictive hotspot maps with confidence intervals	Ebola Virus Hotspot Prediction

Technical Assessments

For relatively accurate predictions in Sentiment Analysis, the challenges around model selection are significant, including the following:

- Evaluation set construction: A standardized test set must be established, covering common cases and edge cases with human annotations, in order to evaluate the accuracy of different models.
- Model selection: Sentiment models are typically trained on predefined datasets such as tweets; differences in context can significantly affect model performance. In

addition, sentiment models come with predefined labels (e.g., happy, angry) that may not align with actual requirements. A recommended approach is to use an LLM for classification. For volumes at the 100k level, the cost is not prohibitive [1,2].

- Predictive insights are highly dependent on domain knowledge, requiring deep involvement from domain experts at every stage:
    - Data collection: Identifying what data is useful for making judgments. In infectious disease prediction, for example, different transmission mechanisms may call for different types of data.
    - Feature engineering: Injecting prior knowledge to help the model fit more efficiently is a practical approach, especially with smaller datasets (e.g., effectively constraining the parameter space). Examples include assigning higher weights to more recent data, or combining two different signals such as CTR as a representative feature.
    - Performance evaluation and calibration: The ability to calibrate both data and models based on observed results when comparing the outcomes of different models.
- Using generative models can reduce reliance on domain experts, as distribution parameters can be designed based on historically similar events.
- Responsive design and performance: Taiwanese readers consume news heavily on mobile devices, making the interaction experience and loading speed of interactive visualizations on mobile a practical challenge that must be addressed.

1. Model Pricing Estimator: Estimate model costs based on input tokens, output tokens and number of api calls.
2. The price is low for the following reasons:
    1. The number of output tokens is very low, so the cost remains minimal. Even with the reasoning feature enabled, it only requires about 50 tokens, though this depends on reasoning length.
    2. Since it's a relatively simple task, smaller models can be used.
    3. If we use batch mode, the cost can be reduced by an additional 50%.

Fact Checking

Integrating AI into the content moderation workflow for fact-checking and source verification, ensuring information accuracy before publication and enhancing reader trust.

Prerequisites

1. Article Content & Metadata
    a. Draft or published articles pending verification
    b. Headlines, publication date, author, and category
2. External Structured Data
    a. Established fact databases and reference repositories
    b. Trusted source whitelists (e.g., verified accounts, specific topic tags)
3. External Unstructured Data
    a. Social media posts (e.g., Twitter/X content for breaking news monitoring)
    b. Third-party news sources and wire services
4. Editorial Team Access
    a. Editors available for human-in-the-loop confidence score review
    b. Subject matter experts for edge-case verification

Modelling Processes & Model Outputs

Use Case	Required Data	Modelling Process	Model Output	Reference Case
Automated Fact-Checking	1 + 2 + 4	Entity extraction from article text; comparison against fact databases; confidence scoring with human-in-the-loop review	Fact-check annotations with confidence scores appended to articles	Checkmate Fact-Checking Tool
Breaking News Verification	3 + 4	Whitelist-based spam filtering (tag or account selection); event detection; AI-assisted claim verification	Verified or flagged social media claims with source links	Twitter/X Breaking News Fact-Checking

Technical Assessments

Data source stability: Social media platforms impose access restrictions, making consistent retrieval of content a persistent challenge. Whitelist-based approaches — such as filtering by verified accounts or specific hashtags — are more practical than broad-scope crawling.

Fact database currency: Reliable fact databases may not be updated in real time. For time-sensitive breaking news, real-time claim verification may still require manual editorial review, as automated cross-referencing against outdated databases can produce misleading confidence signals.

Agentic verification pipeline: Implementing an agentic flow can automate the inspection and verification process end-to-end, extracting key claims and gathering potential evidence, then surfacing only the items requiring human judgment to editors. This reduces manual workload while preserving editorial oversight and accuracy.

Existing user-status dependency: Current AI fact-checking approaches tend to rely on existing user reputation signals (e.g., account credibility scores) for claim validation. This introduces a dependency on platform-specific metadata that may not be consistently available or transferable across sources.

Content Optimisation

Adjusting content based on reader motivations, behaviours, and preferences to ensure more relevant and valuable information is delivered at the individual reader level.

Prerequisites

1. **Article Content & Metadata**
    a. Full text of past articles, headlines, tags, publication date, author, category
2. **Reader Behaviour Data**
    a. Click-through records, dwell time, reading patterns, engagement metrics (e.g. shares, comments)
    b. Sufficient historical volume to move beyond cold-start; LLMs can partially mitigate early-stage data scarcity
3. **User Needs Classification Framework**
    a. A defined taxonomy for categorising reader intent and needs (e.g. Ringier-style user-needs model)
    b. Must be finalised before system build, as it determines annotation standards, training objectives, and evaluation metrics
4. **CMS Integration Access**
    a. Access to the content management system for low-latency inference and recommendation delivery
    b. Ability to display confidence scores and fallback mechanisms to maintain editorial trust
5. **Editorial / Domain Expert Access**
    a. Editorial staff available for framework validation, annotation quality control, and ongoing adoption feedback

Modelling Processes & Model Outputs

Use Case	Required Data	Modelling Process	Model Output	Reference Case
User Needs Content Classification	1 + 3	Framework definition, article annotation, classifier training, user-need labelling, metadata tagging	Articles classified by reader need with auto-generated metadata tags	Ringier User Needs Model
Headline & Metadata Optimisation	1 + 3 + 4 + 5	SEO keyword extraction, click-through and semantic fidelity balancing, confidence scoring	Optimised headline suggestions with confidence scores and consistent metadata tags	Ringier Content Optimisation
Newsletter Personalisation	1 + 2	User profiling from behavioural signals, content-user matching, cold-start mitigation via LLM	Personalised content recommendations per subscriber	*The Times* × Twipe Newsletter Tool

Technical Assessments

Structured understanding of reader intent:

The choice of classification framework directly determines annotation standards, model training objectives, and evaluation methods. However, if the pipeline is designed to be sufficiently modular, the framework itself becomes pluggable, and the cost of replacement and experimentation can be significantly reduced. Investing upfront in architectural flexibility can make later framework adjustments feasible.

Reader behaviour data availability:

The upper bound of personalisation depends on how much user data the media organisation has access to. LLMs can provide a degree of mitigation during the cold-start phase but cannot replace the value of real behavioural data. The core challenge is that reader motivation cannot be measured directly—it can only be inferred indirectly through behavioural signals such as clicks and dwell time. The accuracy of these signals requires continuous validation; for example, clicks may reflect clickbait rather than genuine interest, and training models directly on such signals introduces systematic bias.

Tension between relevance and diversity:

Purely optimising for relevance risks creating filter bubbles. The editorial responsibility of media organisations means this objective cannot be measured solely by machine-learning metrics; it must be accounted for at the system-design level.

Editorial adoption and trust:

For AI-generated suggestions to be genuinely integrated into workflows, editors must develop sufficient confidence in the system. Designing confidence-score displays and fallback mechanisms is recommended to prevent rapid erosion of trust caused by unstable recommendations. Classifier version updates can cause distribution drift between historical and new labels, necessitating a version-management strategy.

Publishing & Platform Promotion

Generating keyword-rich headlines to improve search rankings and increase article traffic. Optimising social post timing, language, and content format to align with evolving platform algorithms and maximise social engagement.

Prerequisites

1. **Article Content & Metadata**
    a. Full text of articles, headlines, tags, publication date, author, category
2. **Search Analytics Data**
    a. Google Search Console data (active search intent, keyword rankings) and Google Discover data (passive push logic, click-through rates)
    b. Understanding of the fundamental difference between the two sources is essential, as their headline optimisation objectives can be mutually contradictory
3. **Social Media Platform Access & Analytics**
    a. Posting access and historical engagement metrics (reach, clicks, shares, comments) from target platforms (e.g. Facebook)
    b. Sufficient reach volume per post to support statistically reliable A/B testing
4. **Brand Style Guide**
    a. Documented brand standards and examples of successful headlines to serve as reference for prompt parameterisation
    b. A systematic method for converting qualitative brand criteria into actionable, programmable constraints (scoring rubric)
5. **Experiment & Prompt Version Management Infrastructure**
    a. Version-controlled prompt registry to track iterations, parameter configurations, and associated performance data
    b. Without systematic tracking, mixing prompt versions or diagnosing performance regressions becomes infeasible at scale

Modelling Processes & Model Outputs

Use Case	Required Data	Modelling Process	Model Output	Reference Case
SEO Headline Generation	1 + 2 + 4 + 5	Search intent analysis, keyword extraction, brand-style parameterisation, multi-objective prompt design (Search vs. Discover)	SEO-optimised headline suggestions with brand-compliance scoring	Il Resto × FT Data Science SEO Headline Tool
Social Media Engagement Optimisation	1 + 3	Historical engagement analysis, platform algorithm inference, optimal timing/format recommendation, A/B testing	Optimised social post recommendations (timing, language, format) with engagement predictions	The Telegraph × Echobox Social Optimisation

Multi-Channel Distribution Strategy	1 + 2 + 3 + 4 + 5	Multi-objective optimisation across search and social channels, dynamic feature weighting by target channel, counterfactual analysis for attribution	Channel-specific content variants with performance attribution reports and automated feedback loop	—

Technical Assessments

Publishing and platform promotion encompasses two fundamentally different problems—search optimisation and social distribution—each with independent optimisation logic. While they do not require a unified strategy, both face the same root challenge: the reliability of performance validation.

Conflict of heterogeneous distribution logic (Conflict of Intent):

Search engines (active intent / semantic matching) and recommendation systems (passive push / emotional triggers) have fundamentally opposing optimisation paths. The system must support a multi-objective optimisation architecture capable of dynamically switching the feature weights and constraints for headline generation based on the target channel (Search vs. Discover/Social), rather than relying on a single logic.

Quantifying qualitative standards (Quantifying Qualitative Standards):

Brand style guides and content quality criteria are typically vague, qualitative descriptions that are difficult to translate directly into model parameters. A scoring rubric must be established to convert abstract brand standards into programmable constraints, enabling automated content quality gating. Converting successful headline examples into actionable prompt parameters—rather than mere qualitative descriptions—requires a systematic methodology.

Attribution and statistical robustness (Attribution & Statistical Robustness):

Traffic performance is influenced by multiple collinear factors—article topic, publication timing, and platform algorithm drift—making traditional A/B testing highly prone to false-positive conclusions when sample sizes are insufficient. More precise counterfactual analysis or long-cycle trend modelling is needed to filter background noise and ensure optimisation recommendations are statistically reliable. This challenge is particularly pronounced for small-to-medium-sized media organisations in Taiwan, where traffic volume may not support robust testing.

Engineering governance of the experiment loop (Engineering Governance):

Ad-hoc testing cannot accumulate knowledge, and the lack of version management makes it difficult to trace and scale optimisation outcomes. An experiment lifecycle management system must be established to link prompt iterations, parameter configurations, and back-end performance data into an automated feedback and knowledge-accumulation mechanism. It is also worth noting that off-the-shelf tools

for Discover parameter optimisation are relatively scarce in the industry, with most progress relying on internally accumulated experimental data.

Personalised Recommendations & Interaction

Leveraging reader behaviour and AI to deliver individually relevant content recommendations and interactive experiences, including personalised push notifications and generative AI-powered question-answering over proprietary article archives.

Prerequisites

1. **Article Content & Metadata**
    a. Full text of articles, headlines, tags, publication date, author, category
    b. Content features (topic, format, length) required as initial signals for cold-start recommendation and RAG retrieval
2. **Reader Behaviour & Interaction Data**
    a. Reading history, notification click/dismiss records, bookmarked interests, dwell time
    b. Sufficient interaction volume to move beyond cold-start; for media with limited membership, demographic-based grouping is a more practical starting point than interest-vector clustering
3. **User Profile / Demographic Data**
    a. Basic demographic attributes (age, location, device type) to serve as a fallback segmentation basis during data-sparse phases
4. **Push Notification Infrastructure**
    a. Delivery system with permission management, capable of dynamic frequency control per user
    b. Once notification permission is revoked by a user, the channel is permanently lost; the infrastructure must treat permission retention as a first-class optimisation objective
5. **Vector Search & Retrieval Infrastructure**
    a. A hybrid search stack (vector + keyword) with a reranking layer for the RAG-based chatbot use case
    b. A chunking strategy for news articles that preserves contextual coherence without introducing excessive noise

Modelling Processes & Model Outputs

Use Case	Required Data	Modelling Process	Model Output	Reference Case
Personalised Push Notification Recommendations	1 + 2 + 3	Demographic-based cold-start segmentation, interest-vector clustering as data matures, dynamic push frequency control based on user response history	Personalised article recommendations with individually optimised push timing and frequency	The Hindu Push Notification Personalisation
Generative AI Chatbot (RAG-Based Q&A)	1 + 5	Temporal query parsing, hybrid search (vector + keyword), reranking, LLM generation with	Summary-based answers with verifiable source citations and confidence indicators	Financial Times Generative AI Chatbot

		explicit attribution mechanism, contradiction handling strategy		
Exploration–Exploitation Balanced Recommendations	1 + 2 + 3	Content-feature extraction, bandit-style online learning, dynamic balance mechanism between relevance and diversity to mitigate filter bubbles	Diversified content recommendations with tuneable exploration ratio and engagement feedback loop	—

Technical Assessments

Exploration versus exploitation balance in recommendations:

If personalised recommendations over-rely on historical behaviour, they reinforce existing preferences and narrow the content surface (filter bubble), ultimately reducing long-term engagement. Designing a dynamic balance mechanism between “giving users what they want” and “broadening their horizons” is the most fundamental product and technical trade-off in this scenario.

Cold-start severity in media versus e-commerce:

Media content has an extremely short lifecycle—articles need effective recommendations within hours of publication, yet almost no interaction data is available at that point. This cannot be solved by collaborative filtering alone; content features (topic, author, format) must serve as initial signals, complemented by bandit-style online learning for rapid adaptation. For media organisations with limited membership and interaction volume, demographic-based segmentation is a more stable starting point than interest-vector clustering, which tends to produce unstable results under data sparsity.

RAG reliability ceiling determined by retrieval quality:

In generative Q&A workflows, hallucination is often over-emphasised; in practice, the more common failure mode is “retrieving the wrong passage, upon which the LLM faithfully produces an incorrect answer.” The priority for improving system reliability should be: ensure retrieval precision first, then optimise generation. Hybrid search (vector plus keyword) and a reranking layer are typically more effective than prompt tuning. Specific retrieval challenges include: poor exact matching of proper nouns and numerical values in vector space; chunking strategies that either lose context (too fine) or introduce noise (too coarse); and the need for keyword-based fallback to hit terms like “TSMC” or “Fed.”

Citation alignment as a trust bottleneck:

Ensuring the model “says A, cites A” does not happen automatically; an explicit attribution mechanism must be designed into the generation stage, and citation-answer correspondence must be independently verified during evaluation. Without this layer, sporadic citation errors will rapidly erode user trust. When conflicting information appears across sources, the system also needs a defined

strategy—whether to favour recency, present multiple viewpoints, or flag the conflict explicitly.

Query understanding and temporal precision:

Converting natural-language time references into precise retrieval filters is deceptively difficult. Relative-to-absolute time conversion ("last three months"), semantically implied time ("the last inflation peak"), and time-decay weighting for financial news all require dedicated handling. Without robust temporal parsing, the retrieval layer returns irrelevant or outdated passages, undermining the entire RAG pipeline.

Push notification as a scarce, non-recoverable resource:

Once a user revokes notification permission, the channel is permanently lost. Push strategy must treat "do not disturb" as an optimisation objective on equal footing with click-through rate, and design dynamic frequency control based on each user's historical response patterns rather than relying on fixed rules. Notification frequency itself needs to be personalised, which loops back to the data-sparsity constraint.

Advertising

AI solutions to improve the utilisation of unsold ad inventory and deliver more relevant ads to the right audiences, thereby increasing overall revenue efficiency.

Prerequisites

1. **User Behaviour & Browsing Data**
    a. Anonymous and authenticated browsing signals: reading topics, dwell time, visit frequency, source/referral paths
    b. A 90-day rolling window is a common baseline; users with shorter histories will have significantly lower segmentation quality
2. **User Identity & Registration Data**
    a. Login status and first-party identity graph to enable cross-session behaviour linkage, especially critical as third-party cookies phase out
    b. Self-reported demographic data (job title, industry, interests); note that completion rates are typically low and high-value users may provide vague information
3. **Ad Inventory & Campaign Performance Data**
    a. Historical fill rates, CPM/CPC by placement, campaign delivery logs, and unsold inventory records
    b. Real-time inventory availability signals for latency-sensitive backfill operations
4. **Ad Tech / DMP Infrastructure**
    a. Data Management Platform capable of receiving audience segments and applying them as targeting filters in ad serving
    b. Integration path between the segmentation model output and the ad server or SSP for real-time decisioning
5. **Creative Asset Pipeline**
    a. Multi-dimensional creative assets (image, copy, video) with version tracking for audience-specific combinations
    b. Quality-control mechanism for AI-generated creatives to ensure visual and tonal consistency across variants

Modelling Processes & Model Outputs

Use Case	Required Data	Modelling Process	Model Output	Reference Case
Audience Segmentation for Ad Targeting	1 + 2 + 4	Behavioural feature engineering (attention proxy), demographic enrichment, audience clustering, DMP segment creation with reach-precision trade-off management	Describable audience segments for ad targeting, pricing negotiation, and custom audience-insight reports	Financial Times Audience Behaviour Segmentation
AI-Driven Creative Optimisation	1 + 3 + 5	Multi-dimensional creative combination generation, audience × creative matching, joint optimisation of	Optimised creative–audience combinations with performance predictions and	MLB × Claritas AI Campaign Optimisation

		placement / content version / timing	explainability annotations	
Unsold Inventory Recovery	1 + 3 + 4	Real-time inventory monitoring, unfilled-placement pattern detection, automated backfill ad assignment with latency-aware triggering	Automated backfill recommendations that re-monetise unsold placements before the impression window closes	Gazeta.pl AI Inventory Recovery

Technical Assessments

For media organisations whose primary revenue model relies on direct-sold placements or agency-mediated deals, control over the ad-serving process is limited. Under this structure, AI's leverage points concentrate on two directions: strengthening placement pricing power and building direct-sales capability. Both share the same foundation—the depth of first-party audience understanding. The strength of audience-data capabilities determines how far either direction can go.

Structural trade-off between precision and reach:

The finer the audience segments, the more creative versions, and the stricter the targeting filters, the smaller the sample size per decision unit becomes. This trade-off recurs across all three case studies and represents a structural constraint of ad AI systems. There is no solution that simultaneously maximises targeting precision and delivery scale—only explicit trade-offs under defined business objectives.

Identity resolution and cold-start under cookie deprecation:

Unauthenticated users cannot have their behaviour linked across sessions, and this problem will worsen as third-party cookies continue to phase out. New users within a 90-day rolling window have insufficient behavioural data, resulting in significantly lower segmentation quality compared to established users. Both issues share the same root cause: the lack of a stable user-identification mechanism. Building a first-party identity graph through registration incentives is a prerequisite for sustainable audience-data capability.

Proxy-label causality in feature engineering:

Attention metrics are composite scores; if signals generated after ad exposure (e.g. ad interaction counts) are mixed into the design, the metric circularly defines what it attempts to predict. The model then learns historical delivery bias rather than genuine content-engagement behaviour, causing new users who have never been exposed to ads to be systematically undervalued. Additionally, self-reported demographic data suffers from low completion rates, and high-value users tend to provide vague information—meaning feature quality is paradoxically worst for the most valuable audience segments.

Combinatorial explosion and statistical sparsity in creative optimisation:

Multi-dimensional creative combinations (image × copy × video × audience segment) grow explosively and exceed the capacity for manual review. When AI-generated creatives are introduced, visual-style and tonal consistency across versions is

difficult to verify automatically. With fine audience segments and many creative variants, exposure per audience–creative combination becomes sparse, and statistical signals are insufficient—systems may converge prematurely on suboptimal solutions. Using click-through rate or short-term conversion as objectives also drives the system toward sensationalist creatives that diverge from the advertiser's true KPIs (retention, brand favourability).

Explainability and compliance friction in automated decisions:

When placement, content version, and delivery timing are optimised jointly, the three decision variables interact with one another; optimising them separately ignores interaction effects, while joint optimisation significantly increases computational cost. Black-box decision-making also lacks explainability, creating friction in client communication and regulatory compliance reviews.

Real-time latency sensitivity in inventory recovery:

Real-time data analysis and pattern recognition for unsold inventory are sensitive to pipeline latency; if detection of unfilled placements is delayed too long, the remediation mechanism activates after the impression window has already closed. Furthermore, this backfill process is typically decoupled from upstream audience segmentation and dynamic creative optimisation outputs, meaning backfill ads have lower personalisation than normal delivery—long-term accumulation may affect the comparability of overall ad-performance metrics.

Subscription Growth

Machine learning can help optimise the timing, content, and presentation of paywalls while identifying optimal pricing strategies to improve conversion rates.

Prerequisites

1. **User Behaviour & Session Data**
    a. Page-view sequences, article read counts, dwell time, visit frequency, referral sources, and device type
    b. Behavioural sequences of sufficient length for reliable intent inference; new or low-frequency visitors will have significantly lower prediction accuracy
2. **Subscription & Churn Records**
    a. Historical subscription events (sign-up date, plan type, renewal/cancellation timestamps) to serve as positive labels for propensity modelling
    b. Expect severe class imbalance: subscribers represent an extremely small fraction of total users, requiring careful label design and sampling strategies
3. **A/B Testing Infrastructure**
    a. Randomised experiment framework capable of assigning users to control/treatment groups for paywall timing and pricing variants
    b. Essential for causal validation; without randomised experiments, the system cannot distinguish whether paywall timing drives conversion or merely correlates with pre-existing purchase intent
4. **Paywall & CMS Integration**
    a. Ability to dynamically control paywall display rules (trigger point, frequency, messaging) from model output in real time
    b. Integration with the subscription management system to feed conversion and retention outcomes back into the model loop
5. **Legal & Compliance Review for Dynamic Pricing**
    a. Pre-assessment of regulatory requirements in target markets regarding personalised pricing (price discrimination laws vary by jurisdiction)
    b. Awareness that users can trivially compare prices with one another, creating trust risk even where dynamic pricing is legally permissible

Modelling Processes & Model Outputs

Use Case	Required Data	Modelling Process	Model Output	Reference Case
Smart Paywall Timing	1 + 2 + 3 + 4	Propensity scoring with class-imbalance handling, real-time session intent inference, A/B-validated trigger-point selection, conversion–churn trade-off modelling	Per-user paywall trigger rules (timing, frequency, messaging variant) with causal conversion-lift estimates	The Globe and Mail AI Paywall Optimisation

Dynamic Subscription Pricing	1 + 2 + 3 + 5	Price-elasticity estimation, willingness-to-pay modelling, fairness-constrained optimisation, randomised price experiments	Personalised pricing recommendations with compliance guardrails and trust-risk flags	Bild ML Dynamic Pricing
Churn Prevention & Retention Intervention	1 + 2 + 3 + 4	Survival analysis for time-to-churn modelling, sequence models for pre-churn signal detection, uplift modelling for intervention effectiveness, A/B-validated discount offers	Churn-risk scores with recommended intervention actions (e.g. renewal discounts) and causally validated lift estimates	—

Technical Assessments

Short-term conversion versus long-term retention trade-off:

Both paywall timing and pricing optimisation easily fall into the trap of targeting short-term conversion rates. A high conversion rate does not guarantee a high retention rate; overly aggressive intervention strategies may boost subscription numbers in the short term while accumulating long-term user attrition. This trade-off cannot be resolved automatically by models—it requires explicit definition at the business-objective level, incorporating retention and lifetime value alongside conversion as joint optimisation targets.

Personalisation versus fairness—a structural tension:

Whether through dynamic paywalls or dynamic pricing, the higher the degree of personalisation, the more visible the differential treatment of different users becomes. This creates sustained friction across two dimensions: legal compliance (price-discrimination regulations vary by jurisdiction) and user trust (the cost for users to compare prices with one another is extremely low). Critically, this tension intensifies rather than naturally resolves as system precision improves.

Causal identification—correlation versus intervention effect:

A system may observe that showing a paywall after the third article yields the highest conversion rate, and set this as the trigger rule. However, this correlation rests on an unverified assumption: that the paywall caused the subscription, rather than that users willing to read three articles already had higher purchase intent. Without causal identification, the model learns historical delivery bias and continuously reinforces a potentially ineffective intervention point. Resolving this requires work on two fronts simultaneously.

Validating whether the intervention is effective:

A/B testing is the gold standard—randomly assigning users with identical characteristics into groups is the only way to isolate the user's inherent purchase intent from the paywall intervention's effect. When sample sizes are insufficient or

randomisation is infeasible, uplift modelling offers a second-best alternative, though it still requires experimental data as a training foundation and cannot fully replace A/B testing.

Determining when to intervene:

Propensity models, survival analysis, and sequence models address different aspects of timing prediction: propensity models estimate a user's current behavioural inclination; survival analysis models the time distribution from subscription to churn; sequence models identify pre-churn signals from behavioural sequences. Their common application is proactive intervention when the system detects rising churn risk—for example, offering a renewal discount before the user cancels. The effectiveness of such interventions must loop back to causal validation to confirm whether the discount genuinely reduced churn.

User Retention & Lifetime Value

Using AI to identify subscribers at risk of churn and recommend more relevant content or products, further enhancing user lifetime value (LTV).

Prerequisites

1. **User Behavioural Data**
    a. Reading history, click patterns, session duration, page views, engagement frequency
    b. Temporal behavioural sequences for survival analysis (login cadence, reading trends over time)
2. **Subscription & Transaction Data**
    a. Subscription status, plan type, payment history, renewal dates
    b. Historical churn events with timestamps (for model training and class imbalance handling)
3. **Content Interaction Data**
    a. Articles read, comments posted, columnist follows, content shares
    b. Action-level logs with natural occurrence rates (for action probability estimation)
4. **Domain Expert Access**
    a. Subject matter experts for defining meaningful churn signals and validating model outputs
    b. Business stakeholders for LTV component decomposition and calibration of action value scores

Modelling Processes & Model Outputs

Use Case	Required Data	Modelling Process	Model Output	Reference Case
Churn Prediction	1 + 2	Survival analysis, sparse signal extraction, imbalanced sample handling, temporal risk windowing	Churn probability scores with dynamic risk windows	Financial Times Churn Prediction
Next Best Action Recommendation	1 + 2 + 3	Action probability estimation, action value scoring, multi-objective ranking fusion	Personalised action recommendations ranked by expected value	Financial Times Next Best Action
LTV Optimisation	1 + 2 + 3 + 4	LTV decomposition into retention probability, future payment propensity, and cross-sell opportunity;	Composite LTV estimates per user segment with actionable levers	Financial Times LTV Framework

		component-wise modelling and synthesis		

Technical Assessments

Signal Sparsity & Temporal Structure

Retention prediction is fundamentally a survival analysis problem, not a standard binary classification task. User churn risk changes dynamically over time, and different users have varying "hazard windows." Churn events are inherently rare, creating extreme class imbalance, while the effective behavioural signal window before churn is narrow, structurally limiting both the sample size and signal density available for model learning.

Unobservable External Churn Factors

A portion of user churn is driven by external factors (competitor emergence, changes in personal financial circumstances, life events). These users may exhibit completely normal on-platform behavioural sequences before churning, making it impossible for any internal signal to predict such churn. This represents a fundamental prediction ceiling unique to this scenario.

Action Probability Estimation Challenges

Probability estimation relies on historical behavioural data, but certain actions (e.g., commenting, following columnists) have inherently low natural occurrence rates, resulting in sparse positive samples and unreliable probability estimates. This directly impacts the quality of Next Best Action recommendations.

Multi-Objective Score Fusion

Action probability and action value are two fundamentally different scores, and there is no single correct way to combine them. The ranking logic between high-probability, low-value actions and low-probability, high-value actions differs entirely depending on the fusion design, producing completely different recommendation outcomes that are difficult to validate using a single metric.

LTV Optimisation Target Ambiguity

LTV is a composite metric simultaneously influenced by retention probability, future payment propensity, cross-sell opportunities, and word-of-mouth propagation. Modelling only retention probability may produce a hidden bias of "retaining high-risk users at the expense of upgrade opportunities for low-risk users." A more complete framework should decompose LTV into independently modellable components, estimate each separately, and then synthesise, rather than directly substituting behavioural proxy metrics (e.g., clicks, reading duration) for LTV. This gap becomes more pronounced when the correlation between behavioural activity and payment propensity is unstable.